



FACULTY OF ENGINEERING
UNIVERSITY OF RAJSHAHI

RUP - 50,000/C.S-1214

Nº 003556

Class Test

Total Marks:

Roll Number/ Student ID No

Course Code

Date of Exam.....

Signature of Invigilator

Discuss briefly about optimization problem and historical development of it.

Solution:- প্রথমত আমরা optimization এর definition থেকে গড়তে হবে।

Historical Development

Existence of optimization methods can be traced to the days of Newton, Lagrange and Cauchy. Development of differential calculus methods of optimization was possible because of the contributions of Newton and Leibnitz to calculus. Foundations of calculus of variations, dealing with the minimizations of functions, were laid by Bernoulli, Euler, Lagrange and Weierstrass. The method of optimization for constrained problems which involve the inclusion of unknown multipliers became known by the name of its inventor Lagrange. Cauchy made the first application

of the steepest descent method to solve unconstrained optimization problems. In recent high-speed digital computers, made implementation of the complex optimization procedures possible and stimulated further research on newer methods.

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Define Unconstrained Optimization Problem. Write down necessary and sufficient condition for optimality of it.

Solution:-

Definition লক্ষ্য, ,

2nd Part:-

Consider the problem of minimizing the function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ where f is twice continuously differentiable on $\mathbb{R}^n$

$$P \text{ minimize } f(x), x \in \mathbb{R}^n$$

Then the first and second order necessary and sufficient conditions for optimality are as follows:

First-Order necessary conditions for optimality

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a point $x \in \mathbb{R}^n$. If x is a local solution to the problem P , then $\nabla f(x) = 0$

Second-order optimality conditions

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be twice differentiable at the point $x \in \mathbb{R}^n$.

Necessity: If x^* is a local solution to the problem P , then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semi-definite.

Sufficiency: If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite, then there is an $\alpha > 0$ such that
$$f(x) \geq f(x^*) + \alpha \|x - x^*\|^2 \text{ for all } x \text{ near } x^*.$$

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Discuss steepest Descent Method and write down algorithm of it.

Steepest Descent Method

The method of steepest descent is a gradient algorithm where the step size

α_k is chosen to achieve the maximum amount of decrease of the objective function at each individual step. Specifically, α_k is chosen to minimize $\phi_k(\alpha) = f(x^k - \alpha \nabla f(x^k))$. In other words

$$\alpha_k = \arg \min_{\alpha \geq 0} f(x^k - \alpha \nabla f(x^k)).$$

~~The~~ 2nd Part

The steepest descent algorithm proceeds as follows: Suppose that we are given a point x^k . To find the next point x^{k+1} , we start at x^k and move by an amount $-\alpha_k \nabla f(x^k)$ where α_k is a positive scalar called the step size. The above procedure leads to the following iterative algorithm

$$x^{k+1} = x^k - \alpha_k \nabla f(x^k).$$

A typical sequence resulting from the method of steepest descent is depicted in the following figure

